

Educator's Speak

How technology empowers teaching?

Today, engineering graduates need skills to analyse, synthesise and design increasingly complex engineering systems. Practically, the design process follows a hierarchy in which systems are designed first through functional block diagrams, followed by circuits and subsystems to meet the design specifications using circuit/system simulation/modeling, integration, testing and verification tools to end up with a cost-effective end product.

For electrical and electronics system design, available design and simulation software like FPGA Advantage (from Mentor Graphics), Blast RTL and Blast Logic (from Magma), NCSim and RTL Compiler for digital system synthesis (from Cadence), SPICE, LabView (from National Instruments), MATLAB and PSCAD/EMTDC/ETAP 6.0 for power electronics are very useful to ensure the

design objective with optimised design parameters. Integration of these tools in teaching empowers teaching.

Initiatives taken

Electronic design automation software facilitates learning of the operating concepts of electronic devices, circuits and systems and helps the students to understand the design and development issues involved in electronics system design. At BITS, Pilani, the students of electrical and electronic disciplines are exposed to extensive use of such software as SPICE, MATLAB and LabView. In most of the courses, course instructors use information technology tools for instruction and evaluation. In addition to hardware experimental laboratory training, students are exposed to online software design assignments submission and evaluation schemes.



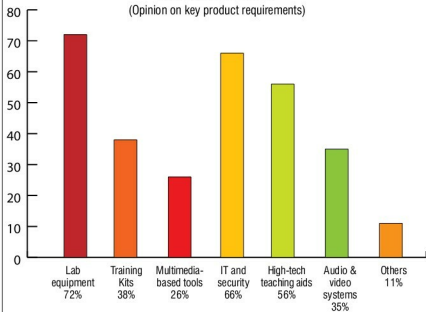
L.K. Maheshwari, vice chancellor, BITS, Pilani

Sufficient exposure to state-of-the-art design and automation tools puts our students at par with other students competing for the industrial/research assignments in the area of VLSI, microelectronics and microprocessor-based embedded system design.

Selection criteria for technology

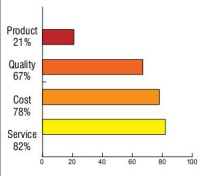
BITS has a strong collaboration with the industry that provides ample opportunities to BITS faculty and students to interact with industry professionals for challenging design and development projects. Interaction with researchers from R&D labs enables us to update our labs and course curriculum to make our students suitable for industry and research assignments. Our strong BITS alumni also helps the department to upgrade the laboratory and software facilities to meet the technology advancements.

Products in Demand
(Opinion on key product requirements)



Unique Selling Points

(Opinion on the focus area for selling)



features seem to be of relatively low importance.

Training a future

Companies today more than ever before realise the need of quality